

Letter Frequency Attack on Monoalphabetic Substitution Cipher

16. Write a C program that can perform a letter frequency attack on any monoalphabetic substitution cipher without human intervention. Your software should produce possible plaintexts in rough order of likelihood. It would be good if your user interface allowed the user to specify “give me the top 10 possible plaintexts.”

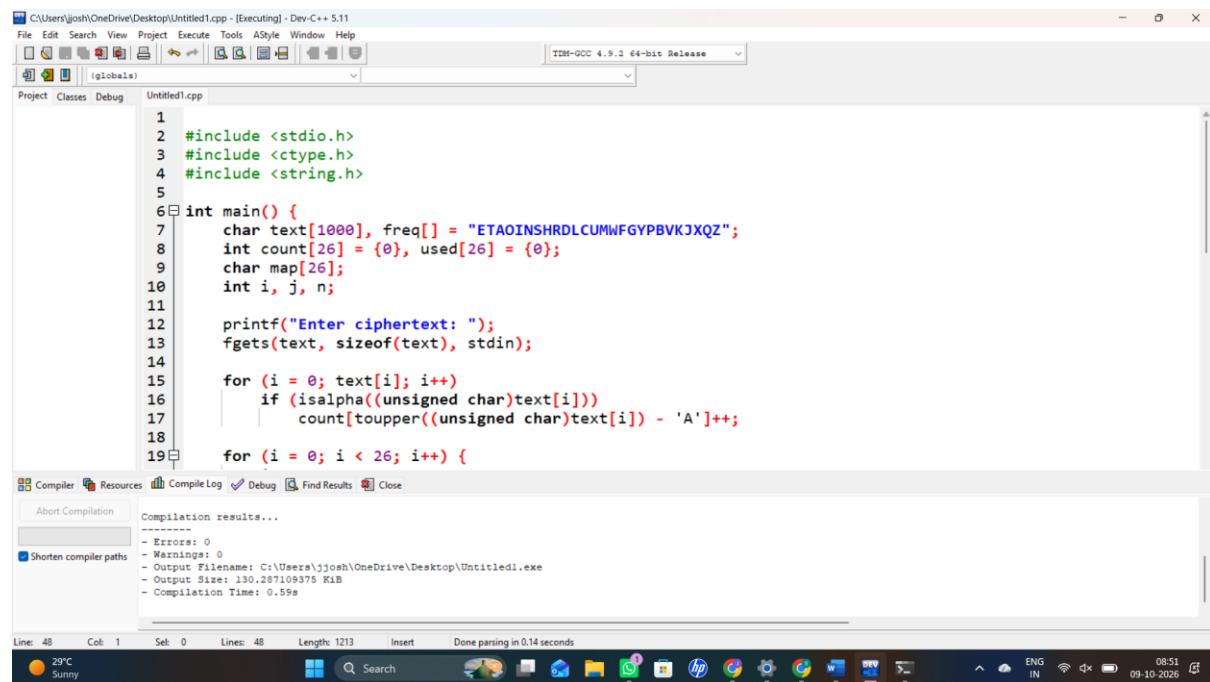
Aim:

To write a C program that performs a letter-frequency attack on a monoalphabetic substitution cipher and displays possible plaintexts.

Algorithm:

1. Start the program.
2. Read the ciphertext from the user.
3. Count the frequency of each alphabet.
4. Compare the frequency order with common English letter frequencies.
5. Replace ciphertext letters with probable plaintext letters.
6. Display the possible plaintext and stop.

Program :



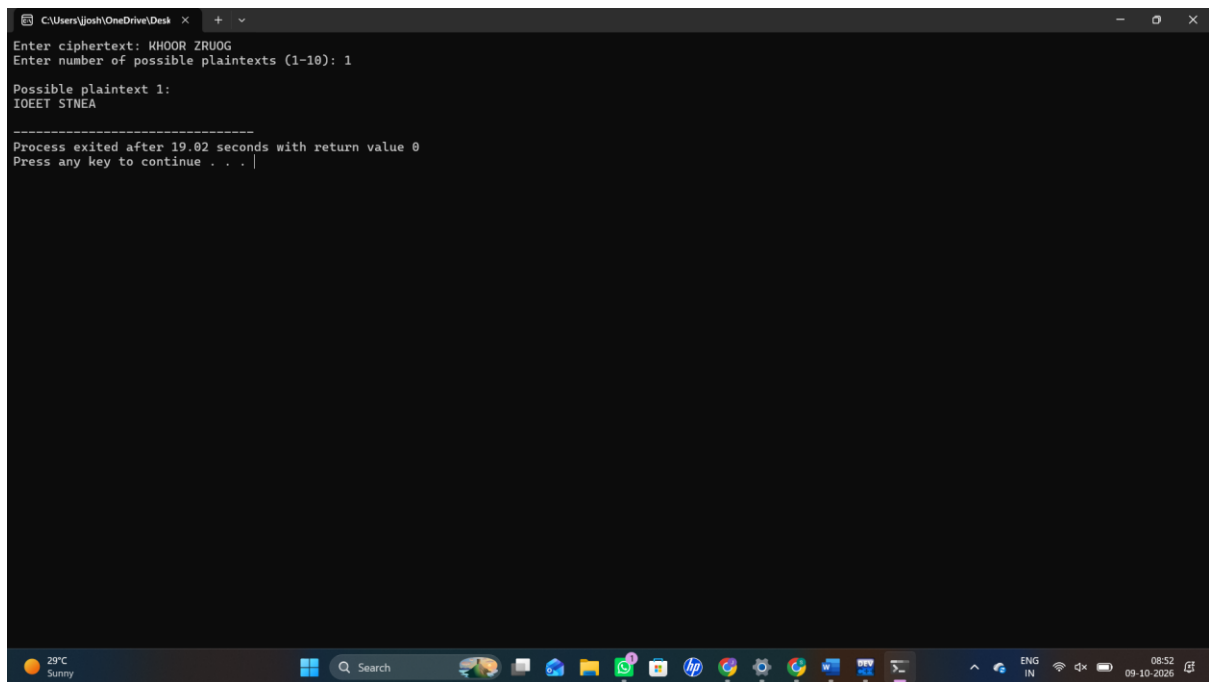
```
1
2 #include <stdio.h>
3 #include <ctype.h>
4 #include <string.h>
5
6 int main() {
7     char text[1000], freq[] = "ETAOINSHRDLCLUMWFGYPBVKJXQZ";
8     int count[26] = {0}, used[26] = {0};
9     char map[26];
10    int i, j, n;
11
12    printf("Enter ciphertext: ");
13    fgets(text, sizeof(text), stdin);
14
15    for (i = 0; text[i]; i++)
16        if (isalpha((unsigned char)text[i]))
17            count[toupper((unsigned char)text[i]) - 'A']++;
18
19    for (i = 0; i < 26; i++) {
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled1.exe
- Output Size: 130.287109375 KiB
- Compilation Time: 0.59s

Line: 48 Col: 1 Sel: 0 Lines: 48 Length: 1213 Insert Done parsing in 0.14 seconds

Input/Output:



```
C:\Users\josh\OneDrive\Desktop >
Enter ciphertext: KHOOR ZRUOG
Enter number of possible plaintexts (1-10): 1

Possible plaintext 1:
IOEET STNEA

-----
Process exited after 10.02 seconds with return value 0
Press any key to continue . . .
```

Result:

The program counts letter frequencies and generates a probable substitution based on common English letter frequencies.